Department of Mathematics and Applied Mathematics

Module 3MS: Metric Spaces 2024

Class test 1 solutions **Total Marks: 40**

1. Let d_E be the Euclidean metric on $\mathbb{R}$ and d_D the discrete metric on $\mathbb{R}$.
   Let $A = \{-1\} \cup (0,4) \cup (4,7]$.

   (a) $\text{int}(A) = (0,4) \cup (4,7)$.

   (b) $\overline{A} = \{-1\} \cup [0,7]$.

   (c) $\text{bd}(A) = \{-1, 0, 4, 7\}$.

   (d) $\text{int}(A) = A$. [4]

2. Suppose (X, d) is a metric space. Define ρ by:

$$\rho(x,y) = \min\{1, d(x,y)\} \text{ for all } x, y \in X$$

   (a) For $x, y, z \in X$:
   Clearly $\rho(x,y) \in \mathbb{R}$.
   (M1) $\rho(x,x) = \min\{1, d(x,x)\} = 0$ since $d(x,x) = 0$. Conversely, if $\rho(x,y) = 0$, then $\min\{1, d(x,y)\} = 0$, so $d(x,y) = 0$ and then $x = y$.
   (M2) $\rho(x,y) = \min\{1, d(x,y)\} = \min\{1, d(y,x)\} = \rho(y,x)$.
   (M3) We show $\rho(x,z) \leq \rho(x,y) + \rho(y,z)$.
   Case 1: If $\rho(x,y) = 1$ or $\rho(y,z) = 1$ (or both), then $\rho(x,z) \leq \rho(x,y) + \rho(y,z)$, because $\rho(x,z) \leq 1$ and $\rho(x,y) \geq 0$ and $\rho(y,z) \geq 0$.
   Case 2: Otherwise, $\rho(x,y) = d(x,y)$ and $\rho(y,z) = d(y,z)$.
   So $\rho(x,z) \leq d(x,z) \leq d(x,y) + d(y,z) = \rho(x,y) + \rho(y,z)$.

   (b) $B(0; \frac{1}{2024}) = (-\frac{1}{2024}, \frac{1}{2024})$

   (c) $B(0; 2024) = \mathbb{R}$ [8]

3. (a) True.
   Let (X,d) be a metric space, for A,B subsets of X, we have $A \subseteq A\cup B$ and $B \subseteq A \cup B$, therefore $\overline{A} \subseteq \overline{A\cup B}$ and $\overline{B} \subseteq \overline{A\cup B}$, from which we have $\overline{A}\cup\overline{B} \subseteq \overline{A\cup B}$. Now, let $x \in \overline{A\cup B}$, then there exists a sequence (x_n) in $A\cup B$ such that $x_n \to x$ as $n\to\infty$, we then have that either A or B contain infinitely many elements of sequence (x_n), suppose without loss of generality that it is A, then we have that A contains a subsequence (x_{n_k}) of (x_n) and $x_{n_k} \to x$ as $n \to \infty$ and so $x \in \overline{A} \subseteq \overline{A}\cup\overline{B}$.

   (b) False.
   Consider $(\mathbb{R}, d_E)$. Let $A = (0,4]$ and $B = [4,7)$, we have $\text{int}(A\cup B) = (0,7)$ while $\text{int}(A)\cup\text{int}(B) = (0,4)\cup(4,7)$.

   (c) False.
   Consider $(\mathbb{R}, d_E)$. Let $A = \mathbb{Q}$, we have $\overline{A} = \mathbb{R}$ and so $\text{int}(\overline{A}) = \mathbb{R}$ while $\text{int}(A) = \emptyset$.

   (d) False.
   Consider $(\mathbb{R}, d_E)$. Let $A = \mathbb{Q}$, we have $\text{bd}(A) = \mathbb{R}$ while
   $\text{bd}(\text{bd}(A)) = \text{bd}(\mathbb{R}) = \emptyset$.

[3+3+3+3=12]

4. (a) For each $n \in \mathbb{N}$, apply the condition in line (3) using $r = \frac{1}{n}$. Since $B(a, \frac{1}{n}) \not\subseteq A$, there exists an element x_n of $B(a, \frac{1}{n})$ that is not in A.

   (b) For each $n \in \mathbb{N}$, $0 \le d(x_n, a) < \frac{1}{n}$, since $x_n \in B(a, \frac{1}{n})$. Taking limits as $n \to \infty$ gives $d(x_n, a) \to 0$, i.e. $x_n \to a$.

   (c) In line (4) we chose $x_n \notin A$, so $x_n \in A'$.

   (d) By assumption, A' is closed. Since x_n is a sequence in A' and $x_n \to a$, $a \in A'$ also.

   (e) In line (7) $a \in A'$, but in line (3), $a \in A$. [3+3+2+2+2=12]

5. (a) False.
   Consider $(\mathbb{R}, d_E)$. We have $d_E^2(1,3) = 2^2 = 4$,
   but $d_E^2(1,2) + d_E^2(2,3) = 1^2 + 1^2 = 2$, so the triangle inequality fails.

   (b) It is clear that $F \subseteq \bigcap\{K \in P(X) \mid K \text{ is a closed subset of } X \text{ and } F \ll K\}$, because $F \ll K$ implies that $F \subseteq K$.
   For the other inclusion, we show that if $x \notin F$, then
   $x \notin \bigcap\{K \in P(X) \mid K \text{ is a closed subset of } X \text{ and } F \ll K\}$.
   So suppose that F is a closed subset of X and $x \notin F$.
   Then $\text{d}(x,F) = \inf\{d(x,y) \mid y \in F\} = r > 0$.
   Let $K = \{z \in X : d(x,z) \ge \frac{r}{2}\}$.
   Then $x \notin K$.
   Note that K is closed, since it is the complement of an open ball.
   Let $H = B[x, \frac{r}{2}]$.

Then $F \cap H = \emptyset$ since if $a \in H \cap F$, then $r = \text{dist}(x, F) \leq d(a, x) \leq \frac{r}{2}$; contradiction.
We also have $H \cup K = X$, since, for all $a \in X$, $d(a, x) \geq \frac{r}{2}$ or $d(a, x) \leq \frac{r}{2}$.

[2+2=4]